CAPSTONE PROJECT REPORT

AI RESUME AND LINKEDIN PROFILE OPTIMIZER

Value-Added Course (VAC) – OneYes

Student Name	_____
Register Number	_____
Department / Year	_____
College Name	_____
Project Guide / Coordinator	_____
Submission Date	_____

Report draft: complete the implementation, add genuine screenshots/results, and insert your actual GitHub repository URL before final submission.

1. Abstract

The AI Resume and LinkedIn Profile Optimizer is a proposed web-based application that helps students and job seekers improve the quality of their resumes and professional LinkedIn profiles. The system analyses user-provided profile information and gives suggestions to improve clarity, relevance, skills presentation, and professional wording. It can compare resume content with a job description, identify missing keywords, and provide suggestions for improving a LinkedIn headline, About section, and experience descriptions. The aim is to make career documents clearer and better aligned with a selected job role. The tool is intended to support users; it does not guarantee selection or employment.

2. Introduction

A resume and a LinkedIn profile are important tools for presenting a person's education, skills, projects, and experience to recruiters. Many students find it difficult to write professional summaries, describe projects, select relevant skills, and tailor their resumes to a particular job. This project proposes an AI-assisted solution that reviews the information entered by a user and provides practical improvement suggestions. The user remains in control and can review or edit every suggestion before using it.

3. Problem Statement

Students and fresh graduates may have difficulty preparing resumes and LinkedIn profiles that clearly communicate their abilities. Their documents may contain unclear descriptions, repeated information, missing role-related keywords, or weak project summaries. Manually reviewing and improving these documents can take time. Therefore, a simple application is proposed to analyse the content and provide structured, understandable recommendations.

4. Objectives

- To provide a simple interface for entering or uploading resume text.
- To analyse resume sections such as education, skills, projects, and experience.
- To compare resume content with a user-provided job description.
- To identify relevant keywords that may be missing from the resume.
- To suggest improvements for a LinkedIn headline, About section, and experience descriptions.
- To help users create clearer and more professional career documents.
- To keep the user responsible for checking accuracy and approving final edits.

5. Scope of the Project

The proposed system focuses on resume text analysis and LinkedIn profile content suggestions. It may support students, fresh graduates, and job seekers. The first version can accept pasted text and a job description. Later versions may include PDF/DOCX upload, downloadable reports, multiple job-role comparisons, and additional language support. The system should not request LinkedIn passwords or automatically change a user's LinkedIn account.

6. Technologies Used

Technology	Purpose
Python	Core programming and text-processing logic

Technology	Purpose
Streamlit (suggested)	Web interface for user input and results
Natural Language Processing (NLP)	Text cleaning, keyword extraction, and content analysis
spaCy / NLTK (optional)	Language processing utilities
Scikit-learn (optional)	Text similarity and keyword comparison
Git and GitHub	Version control and project-code hosting

Note: These are suggested technologies for a feasible implementation. In the final report, keep only the tools actually used in your working project.

7. Proposed System Architecture

The application follows a simple input–processing–output workflow:

• User Input: The user pastes resume content and, optionally, a target job description and LinkedIn profile text.

• Pre-processing: The application cleans the text, separates sections, and normalises words.

• Analysis: The system checks section completeness, extracts skills and keywords, and compares resume terms with the job description.

• Recommendation Engine: The system prepares suggestions for wording, relevant keywords, and LinkedIn sections.

• Results: The user sees a summary, improvement suggestions, and a list of possible missing keywords.

Architecture flow: User → Web Interface → Text Pre-processing → Resume / Job Description Analysis → Recommendation Engine → Results Dashboard.

8. Project Modules

Module 1 – Resume Input: Allows the user to paste resume text. File upload can be added if supported by the implementation.

Module 2 – Resume Section Review: Checks whether key sections such as summary, education, skills, projects, and experience are present.

Module 3 – Job Description Matching: Compares the resume with a job description and highlights related and potentially missing keywords.

Module 4 – LinkedIn Profile Optimizer: Provides suggestions for the headline, About section, skills, and experience descriptions.

Module 5 – Improvement Report: Displays findings and suggestions in a clear format for the user to review.

9. Implementation Plan

• Create the project folder and set up the Python environment.

• Build the input page for resume text, job description, and LinkedIn text.

• Implement text cleaning and section detection.

• Implement keyword extraction and text similarity or matching logic.

• Generate clear suggestions based on the analysis.

• Display the results in a readable dashboard.

• Test the application using sample resumes and different job descriptions.

• Upload the working source code and README file to GitHub.

Implementation status: This report describes a proposed project structure. Replace this section with the exact steps, screenshots, libraries, and code details from your completed application.

10. Expected Output

• A resume review showing detected sections and areas that may need attention.

• A list of keywords found in the resume and keywords from the job description that may be missing.

• Suggestions to improve resume bullet points and the professional summary.

• Suggested LinkedIn headline and About-section wording.

• A final set of recommendations that the user can review and edit.

Add screenshots of your actual running application here. Do not use sample screenshots as proof of completed implementation.

11. Advantages

• Provides structured feedback in one place.

• Can save time during the first review of a resume.

• Helps users notice relevant job-description keywords.

• Supports clearer descriptions of skills and projects.

• Can be useful for students and fresh graduates.

12. Limitations and Responsible Use

• AI-generated suggestions may be inaccurate or unsuitable and must be checked by the user.

• The system cannot guarantee interviews, job offers, or recruiter selection.

• Users should not add skills, qualifications, or work experience they do not have.

• Personal information should be handled carefully; avoid storing resumes without permission.

• The application should not ask for LinkedIn login credentials or claim to edit a LinkedIn account unless an approved integration is actually implemented.

13. Testing

Test the application with different sample inputs: a resume with all major sections, a resume missing a skills section, and a resume paired with a relevant job description. Check whether the system identifies sections consistently, reports keyword matches correctly, and displays understandable suggestions. Record the actual test cases and results in the table below.

Test Case	Input / Action	Expected Result	Actual Result
1	Paste a complete resume	Show detected sections	To be filled
2	Paste resume with missing sectidd	entify section needing attention	To be filled

Test Case	Input / Action	Expected Result	Actual Result
3	Add a job description	Show matching and missing keywords	To be filled
4	Enter LinkedIn About text	Display improvement suggestions	To be filled

14. Conclusion

The AI Resume and LinkedIn Profile Optimizer proposes a practical way to help students and job seekers review their career documents. By checking resume sections, comparing content with a job description, and suggesting improvements to LinkedIn profile text, the application can make the preparation process more organised. The project also highlights the importance of truthful information, user review, and responsible handling of personal data. The usefulness of the final system should be demonstrated through its working application and actual test results.

15. GitHub Repository

GitHub Repository URL: _____

Repository name: AI-Resume-and-LinkedIn-Profile-Optimizer

GitHub README should include: project description, features, technologies used, installation steps, how to run the application, screenshots, and project limitations.

Important: Paste your real, publicly accessible (or evaluator-accessible) GitHub repository URL above after uploading your completed code. Do not submit a made-up link.

16. Final Submission Checklist

• Complete and test the individual capstone project.

• Upload the actual project code to GitHub.

• Add a README file with setup and run instructions.

• Add screenshots of the working project.

• Enter student details and the actual GitHub repository URL in this report.

• Export the final report as PDF.

• Upload the PDF and required submission details using the Google Drive link shared by the coordinators.

Date note: The notice you shared lists Wednesday, 21 October 2026 as the submission deadline. That date has already passed as of 30 October 2026, so please confirm the updated deadline with your coordinator.